
1. PROJECT OVERVIEW

This project transforms raw business data into meaningful insights using Microsoft Excel for data-driven decision making.

2. COMPLEX FORMULAS USED (with Examples)

- a) XLOOKUP (Advanced Lookup)
Formula: `=XLOOKUP(A2, Products!A:A, Products!B:B, "Not Found")`
Use: Fetch Product Price from another sheet. Better than VLOOKUP.
- b) INDEX + MATCH (Powerful Combination)
Formula: `=INDEX(C2:C100, MATCH(E2, A2:A100, 0))`
Use: Looks up value in any direction, faster for large data.
- c) SUMIFS / COUNTIFS / AVERAGEIFS (Multi-Condition)
Formula: `=SUMIFS(D:D, B:B, "North", C:C, "Laptop")`
Use: Total sales of Laptop in North Region only.
- d) IFS & Nested IF (Categorization)
Formula: `=IFS(F2>10000,"High", F2>5000,"Medium", TRUE,"Low")`
Use: Auto-categorize customers based on purchase value.
- e) Data Cleaning Formulas
`=TRIM(CLEAN(A2))` - Remove extra spaces & non-printable chars
`=LEFT(A2,5), =RIGHT(A2,3), =MID(A2,2,5)` - Extract text parts
`=TEXT(B2,"mmm-yyyy")` - Format date
- f) Date & Time Analysis
`=YEAR(Date), =MONTH(Date), =WEEKDAY(Date)`
`=DATEDIF(Start, End, "M")`
Use: To analyze monthly trends and seasonality.
- g) Advanced Logic
`=SUMPRODUCT((Region="North")*(Sales))`
`=IFERROR(VLOOKUP(...),0)`

3. DATA ANALYSIS METHODS

Step 1: Data Cleaning

- Removed duplicates (Data > Remove Duplicates)
- Handled blank cells with `IF(ISBLANK())`
- Standardized text with `TRIM, PROPER`

Step 2: Data Transformation (Power Query)

- Imported data, changed data types, merged tables

Step 3: Exploratory Data Analysis

- Created Pivot Tables to summarize Sales by Month, Region, Product
- Used Pivot Charts for quick insights

Step 4: KPI Calculation

- Total Revenue = `SUM(Sales)`
- Average Order Value = `AVERAGE(Sales)`
- Profit % = `(SUM(Profit)/SUM(Sales))*100`
- YoY Growth = `(Current - Previous)/Previous *100`
- Top Product = `INDEX(MATCH(MAX()),`

Step 5: Data Visualization - Dashboard

- KPI Cards with Conditional Formatting
- Slicers for interactive filtering (Region, Product, Year)
- Charts: Bar Chart (Product performance), Line Chart (Monthly Trend), Donut Chart (Region share), Column Chart
- Used Data Bars and Color Scales

4. KEY INSIGHTS (Sample - you can edit as per your data)

- Highest Sales Month: December
- Top Region: North India contributed 38% of sales
- Best Selling Product: Laptop
- YoY Growth: 15% increase

5. CONCLUSION

This project demonstrates proficiency in Excel for real-world business analytics and is completed as part of Yuva Internship Program.

Skills: MS Excel Advanced, Data Cleaning, Dashboard, Business Intelligence